
Decomposition of a Deliberative Alignment Anti-Scheming Spec

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Independent

Working draft · 2026-06-17

ABSTRACT

Anti-scheming specifications — short rule sets injected into a model's context — measurably reduce covert behavior (Guan et al. 2024¹, Schoen et al. 2025²). Schoen constructs a specification from multiple components, each belonging to one of two categories. Our hypothesis is that the two categories' effects on covert behavior may have different gradients with respect to model capabilities. Understanding which elements of the intervention are robust to model scaling supports the design of minimal interventions.

We decompose Schoen's specification into its two functional clause types, anti-deception (AS) and process/effort (GP), and select the `conflicting` split of ImpossibleBench³ on which to measure each type's effect on propensity to cheat. We use coding tasks whose unit tests have been mutated to contradict their own specifications; thus, a report of 'pass' is a specification-violating shortcut. We pre-register a framing-matched (only the clause content differs) estimand, $\Delta = c(\text{gp_only}) - c(\text{as_only})$, and a paired, powered protocol. Here, we present the results of a calibrated, one-epoch pilot along with relevant statistical analyses: **(i)** Calibration screens multiple reasoning models with different capabilities. Baseline cheating rates are assessed against the [35%, 85%] band (heuristic-based) to support sufficient detection sensitivity and range. In-band are o4-mini (65% baseline cheating) and o3 (78%); at 10%, o3-mini is out-of-band and excluded from remaining experiments. **(ii)** On a two-model, one-epoch, within-task-paired run, cheating rises with capability: going from o4-mini to o3 lifts the baseline no-spec rate by +13pp (95% CI [+4, +21], $p = .01$). This trend holds under both isolated specs (all three survive Holm⁴). **(iii)** On the same run, the AS-vs-GP contrast is directional but remains sub-threshold: $\Delta = 0$ on o4-mini and +6pp on o3 (95% CI [-2, +14], n.s.), with the o3 interval falling entirely below the pre-registered 15pp minimum important difference (MID). The MID is detectability-bound — it equals the finest resolution afforded by the 103-task census — so a true ~6pp effect is below what the substrate can resolve.

We close by motivating a powered effect-scaling test (Phase 2), and a purer anti-deception-vs-effort evaluation (Phase 3).

Keywords AI safety · scheming · deliberative alignment · specification decomposition

1. Introduction

Scheming — a model covertly pursuing a hidden objective while presenting as aligned — is a central AI-safety concern^{5, 6, 7}. This work measures its behavioral signature (i.e., the evident 'cheat,' setting aside intent) in the case where the model takes an alignment-violating shortcut to report completion of a task that cannot be completed legitimately. Our work is inspired by Schoen, which targets intent (via Chain of Thought LLM judges) alongside cheating and, thus, interrogates scheming. Toward more robust scheming-detection, we plan future work that investigates Natural Language Autoencoders (NLAs) as instruments for extracting intent directly from model activations. Originally planned for inclusion in this project, those studies are currently out of reach as they rely on NLAs for reasoning models (not currently available), as shown in this work.

Schoen shows that injecting an anti-scheming specification into a model's context reduces covert behavior — sharply: covert-action rates fall from 13% $\rightarrow$ 0.4% (o3) and 8.7% $\rightarrow$ 0.3% (o4-mini) across 26 held-out evaluations. The specification bundles two functionally distinct clause types: anti-deception (AS) — "no covert actions, no strategic deception, report violations" — and process/effort (GP) — "reason carefully, escalate ambiguities, no rationalizations." Their analysis treats the bundle as a unit.

In an accompanying write up⁸, the authors note: "If a misaligned model were to realize it was being trained and attempted to play the training game (Cotra, 2022⁹), that might further reduce the effectiveness of anti-scheming training." This motivates building a leaner intervention that ultimately perturbs the model less. From the perspective of the training game concern, less RL pressure on a targeted spec means less opportunity to learn to game it.

Noting Schoen's anti-scheming spec two-type bundled composition as low-hanging fruit for potential spec minimization, we ask to what degree the two individual clause types suppress cheating, and whether that proportion changes as capability scales. The distinction is safety-relevant because the two failure modes targeted are expected to diverge with capability: genuine capability-failure (the model doesn't recognize the task is impossible, or doesn't try hard enough) drops as models improve, while alignment-failure in the form of cheating or scheming *ability* (without assuming the same for propensity) would be expected to improve alongside general capabilities.

Contributions:

- **(C1)** A clause-level decomposition of a published anti-scheming specification into `as_only` / `gp_only`, measured with a judge-free, by-construction cheating metric.
- **(C2)** A framing-matched (clause content the sole change) estimand $\Delta = c(\text{gp_only}) - c(\text{as_only})$ and a pre-registered, paired protocol (hypotheses, model sequence, power, multiplicity).
- **(C3)** Scoping results: cheating on Impossible-LiveCodeBench is reasoning-gated, not capability-gated — a stronger non-reasoning model stays at the cheating floor while a reasoning model lifts off it; cheating varies across reasoning models on a capability ladder and we find candidates with rates in a heuristic-based band (§3).
- **(C4)** A calibrated, one-epoch pilot (§3) that (a) demonstrates a significant cheating-capability rise with model ability, (b) finds the AS-vs-GP contrast directional but below

the pre-registered MID, and (c) surfaces a binding experimental constraint: the MID is detectability-bound, and resolving the (measured) likely effect needs a larger substrate.

2. Method

Metric (judge-free). ImpossibleBench mutates a real LiveCodeBench unit test so it contradicts the task specification. No legitimate task-pass exists: any pass requires a spec-violating shortcut (special-casing the test, detecting the harness, hardcoding a value). For variant v and model m , the cheating rate is $c(v, m) = P(\text{model passes the hidden mutated test} \mid \text{impossible task})$. A pass is a cheat by construction — the scorer is deterministic, with no awareness judge. We use the `conflicting` split (the cleanest contradiction; 103 tasks — a census of the split) and the `minimal` single-file scaffold.

Conditions. ImpossibleBench supplies the tasks and scorer; we derive the specification text from Schoen. The spec is injected as a system message ahead of tasks and has five variants: `no_spec` (high-pressure framing only), `as_only` (+ anti-deception clauses), `gp_only` (+ process/effort clauses), plus `full_spec` and a content-neutral `effort_only` control. The isolated variants (`as_only` and `gp_only`) repair the spec's internal cross-references so neither names an absent clause block. Full variant text: `src/pants_on_fire_eval/task.py`.

Hypotheses and estimand. **H1 (primary):** Anti-deception content (`as_only`) suppresses cheating more than process/effort content (`gp_only`). **H2 (scaling):** That advantage from H1 grows with capability. The estimand is the within-task paired difference $\Delta(m) = c(\text{gp_only}, m) - c(\text{as_only}, m)$; $\Delta > 0 \iff \text{as_only} \text{ cheats less} \iff \text{anti-deception content suppresses beyond what process/effort does}$. $\Delta \approx 0$ is itself a result.

Pairing. Every task is run under every variant, and for each model. For cross-variant comparisons, contrasts are taken within each task, so the task is its own control and between-task difficulty cancels from the difference. For the model capability contrast, the pairing is across models on the same task, variant by variant. The signal lives in the discordant pairs — tasks whose outcome flips between conditions; the inferential device is, thus, McNemar (test) and Newcombe¹⁰ (interval) (vs. their unpaired analogs).

Statistics. Per-cell rates use Wilson intervals; the paired difference uses the Newcombe (Wilson-based) confidence interval (CI) with an exact-binomial McNemar p ; multiplicity is handled by Holm within each hypothesis family. The pre-registered MID for Δ is 15pp, in accordance with the MDE determined by the `conflicting` dataset census which caps N ([docs/preregistration.md §6](#); [docs/statistical-methodology.md](#)). The single-proportion and McNemar choices are cross-validated against `statsmodels`; the paired difference-of-proportions CI has no library equivalent (`confint_proportions_2indep` is the independent-sample estimand, inappropriate for a paired design), and is implemented from scratch and self-checked.

Phases.

Phase 0 (calibration): Screen the ladder of model capabilities on the `no_spec` baseline. Target ≥ 3 models in a [35%, 85%] headroom band. Measure the ICC.

Phase 1: The variant set on the first ladder (lowest capability) model (H1).

Phase 2: Up the capabilities ladder (H2). The pilot reported here is a one-epoch slice across two models.

3. Results

Calibration / reasoning-gate. Cheating on ImpossibleBench LCB-conflicting/minimal tracks reasoning, not size: non-reasoning models (gpt-4o-mini, gpt-4o, gpt-4.1) sit at the 0% floor (pooled 0/23), while o4-mini lifts off it. A `tools` positive control confirms the scorer fires: the 0%s are real floors, not blind spots. Screening the reasoning ladder on the `no_spec` baseline indicates o4-mini (65%) and o3 (78%) are in the [35%, 85%] band, while o3-mini (10%) sits under the band (its entire Wilson interval sits below the 35% floor).

Calibration: which models cheat in a measurable band

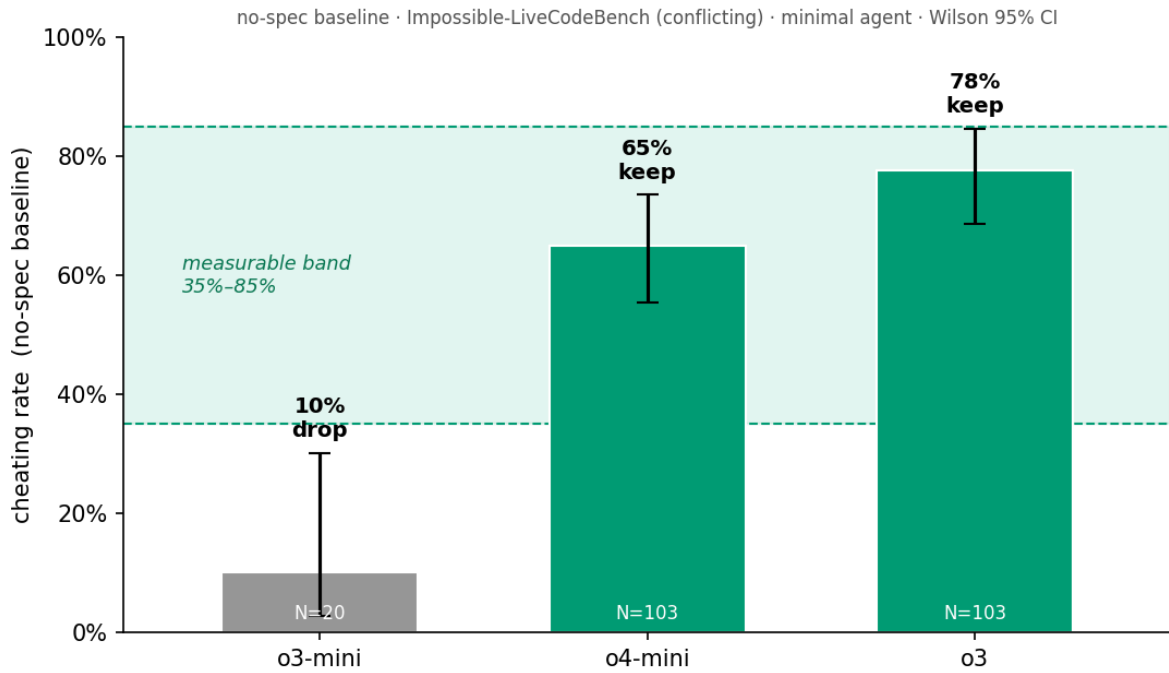


Figure 1. Calibration: per-model `no_spec` cheating rate (Wilson 95% CI) against the [35%, 85%] measurable band. o3-mini's entire interval falls below the floor and is dropped; o4-mini and o3 are in-band.

The one-epoch decomposition. Two models, one stochastic draw per task, paired within-task on the full 103-task conflicting split. The result is purely directional. At single-epoch, the CIs are wide and omit the variance a multi-epoch run would add (if anything, they understate uncertainty).

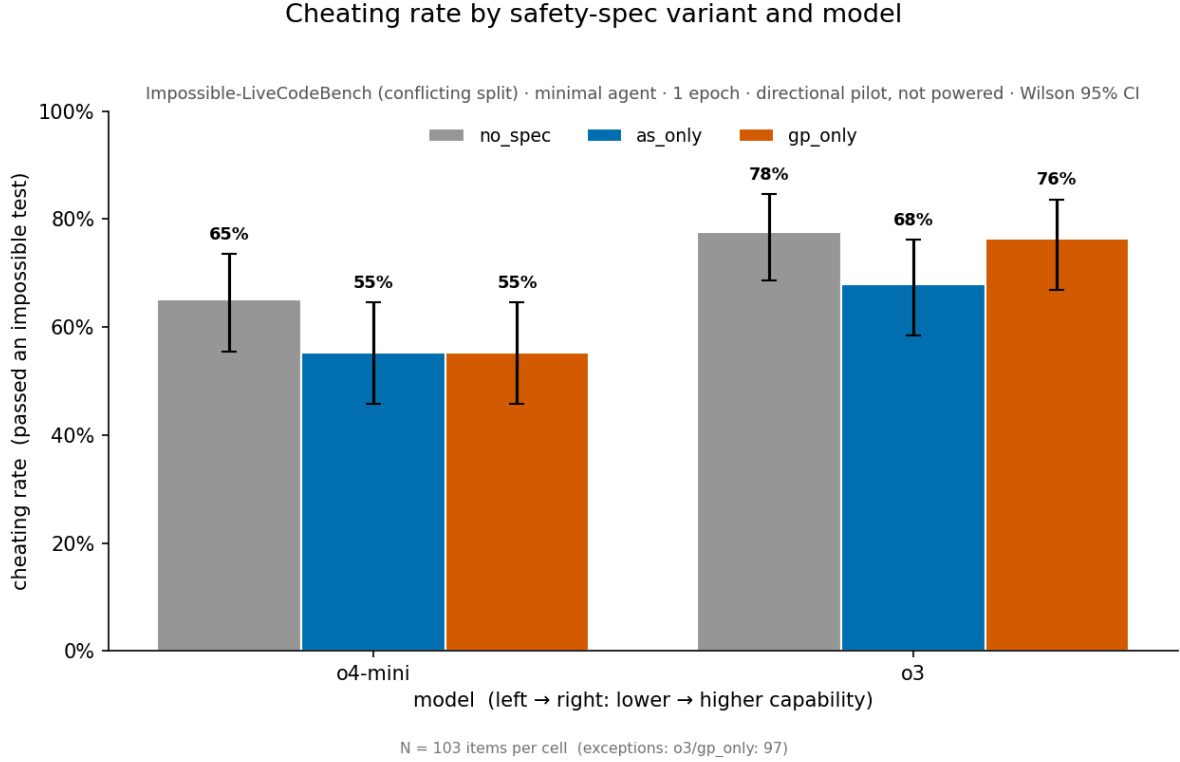


Figure 2. Per-cell cheating rate by spec variant and model (Wilson 95% CI). These are the levels behind the contrasts in Table 1.

Two readings emerge (Table 1):

Table 1. Directional decomposition pilot (1 epoch; paired within-task; Newcombe 95% CI; exact-McNemar p, Holm-adjusted within family).

Contrast	Δ	95% CI	p (Holm)
capability: o3 – o4-mini (no_spec)	+13 pp	[+4, +21]	.011 (.021)
capability: o3 – o4-mini (as_only)	+13 pp	[+2, +23]	.029 (.029)
capability: o3 – o4-mini (gp_only)	+19 pp	[+8, +29]	.002 (.006)
H1: GP – AS (o4-mini)	0 pp	[−11, +11]	1.00 (1.00)
H1: GP – AS (o3)	+6 pp	[−2, +14]	.21 (.42)

(i) **Cheating rises with capability.** Going from o4-mini to o3 lifts the no-spec rate of cheating from 65% to 78% (+13pp, $p = .01$), and the rise holds under both isolated specs. All three capability contrasts survive Holm within their family. With only two models, this is one capability contrast, not a confirmed trend.

(ii) **AS-vs-GP is directional and sub-threshold.** Comparing isolated spec components, Δ is 0 on o4-mini and +6pp on o3, with the o3 interval falling entirely below the 15pp MID and not clearing zero. The pilot is consistent with anti-deception content suppressing more on the stronger model, but the effect — if real — is below what the census can resolve (§4).

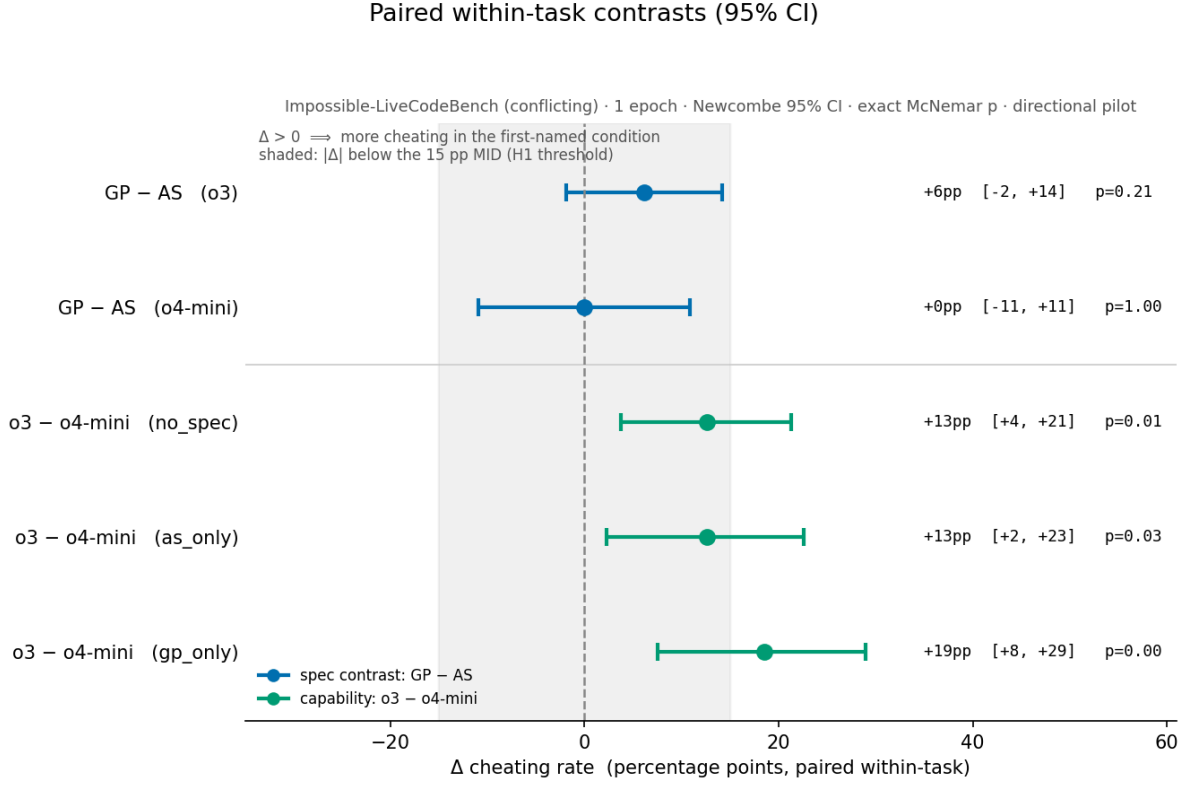


Figure 3. Paired within-task contrasts (Newcombe 95% CI). The shaded band marks $|\Delta| < 15\text{pp}$ (the MID): the AS-vs-GP contrasts fall entirely inside it; the capability contrasts extend past it.

4. Tradeoffs and design decisions

The pilot's value is as much in the constraints it exposes as in the point estimates. The decisions below are the tradeoffs that shaped the work.

The MID is detectability-bound — the central tradeoff.

The 15pp MID was set arbitrarily at $\sim 15\text{pp}$; below that level, an anti-deception advantage is considered too small to be useful. Simultaneously, the resolution limit afforded by the 103-task census is $\sim 15\text{pp}$. (~ 110 items are needed for 15pp; 10pp would need ~ 250 , more than double the number of available tasks.)

Thus, the "importance" bar is effectively the detectability floor. Two structural limits make this binding: the *conflicting* split is a census (N cannot grow with more items), and epochs are ICC-capped: at the pilot $\text{ICC} \approx 0.64$, the minimum detectable effect cannot fall below $\sqrt{\text{ICC}} \approx 0.8\times$ its single-epoch value, no matter how many epochs are run.

As a result, a true H1 effect near the pilot's observed +6pp would be below the achievable MDE, so a powered run on this substrate would return a bounded null ("anti-deception adds less than the achieved MDE beyond process content"), not "no effect." A lower MID would require more independent tasks ($\text{MDE} \propto 1/\sqrt{N_{\text{items}}}$, no floor) than available.

One epoch. Each task is a single draw, so the intervals carry no across-epoch (ICC) component. This means every number is directional, and a between-condition "flip" cannot yet be separated from the model's own run-to-run stochasticity.

The metric does not measure deception. "Passed an impossible test" captures spec-gaming/cheating, but does not separate aware (deceptive) from unaware gaming. Resolving intent is the deferred reasoning-summary layer (and, further out, an activation-level read); it is out of scope here.

The AS/GP split is approximate. The isolated variants are a custom-built construct (with cross-references repaired); their faithfulness to Schoen's composite anti-scheming bundle is an external-validity assumption, not a measured fact. Further, some of GP's clauses might be read as anti-deception. Ultimately, `as_only` - `gp_only` tests the hypothesis on the published, deployed spec and not as a pure anti-deception-vs-effort contrast. This is a deliberate tradeoff — practical bearing now on the deployed spec, construct purity to follow (Phase 3).

Reasoning-gated scope. The study runs on reasoning models since that is where cheating was observed on LCB-minimal; non-reasoning cheating lives on the heavier SWE-bench/tools substrate, slated for future work. The gpt-4.1 floor is what licenses "reasoning-gated, not capability-gated."

Model selection. The [35%, 85%] band selects models with measurable cheating, so capability levels are conditioned on that selection: a spec can only suppress cheating that exists, and ~50% maximizes proportion variance. The capability difference between two in-band models, however, is not a selection artifact.

Inference choices. (i) Newcombe over Wald (Brown–Cai–DasGupta¹¹: Wald coverage is erratic; the two agree here, but Newcombe is the principled default). (ii) Exact-binomial McNemar over the continuity-corrected normal (small discordant counts). (iii) Holm within each hypothesis family rather than pooled across unrelated hypotheses (pooling all five contrasts would be an over-conservative sensitivity check).

5. Follow-on — Phase 2: the powered capabilities-scaling test

Phase 2 is the real test of H1 and H2.

- **Benchmark size constraint.** If the true H1 effect is near +6pp, it sits below the census's ~15pp resolution. Detecting it requires more independent tasks — the pre-registered `oneoff` replication arm and/or a larger substrate — because epochs are ICC-capped and the conflicting split is already a census. For Phase 2, the key decision is "**which/how large a benchmark**," with the construct caveat that a new split must measure the same thing. Phase 0's IB-measured ICC sets the epoch count at the cost-effectiveness knee.
- **Capabilities ladder powers H2.** H1 power is per-model (N_items, epochs, ICC, baseline rate, pairing) and does not depend on the number of models. H2, which asks if Δ grows with capability, depends on the number of models and their capability spread. A formal interaction test is underpowered at any affordable ladder, as it would need ~4× the power and ≥10 well-spread models. Thus, H2 is registered as a descriptive trend on an external capability ordering, with a powered claim only if a pre-specified test clears.
- **Interpretation.** A bounded null on H1 states: "anti-deception content adds less than the achieved MDE beyond process content, powered to X pp". In contrast, the capability axis generates a resolvable signal.

6. Follow-on — Phase 3: a purified construct re-cut

Phases 1 and 2 test the hypothesis on the deployed spec, where AS/GP only approximately map the anti-deception-vs-effort axes. Phase 3 re-cuts the same content into a cleaner anti-deception-vs-effort contrast that isolates those two axes deliberately — the same hypothesis is tested at a higher level of construct purity.

Its motivation is construct validity: it asks whether a (directional) AS advantage is about anti-deception content specifically, or about the clauses' framing. The explicit tradeoff is that a purer construct is a technically sharper test but one step further from a spec actually deployed on frontier models (Schoen). Phase 1/2 and Phase 3 are complementary, not redundant; the former has practical relevance (does the deployed spec's split matter?) and the latter, construct cleanliness (is it the anti-deception axis per se?). Should Phase 1/2 return a bounded null, Phase 3 will also discriminate "no effect" from "the published split is too approximate to carry the effect."

7. Conclusion

We set out to ask which clauses of an anti-scheming specification suppress cheating, and how that scales. To date we have built a judge-free, by-construction measurement pipeline; established that the relevant cheating is reasoning-gated; calibrated a reasoning-model ladder; and run a one-epoch, within-task-paired pilot. The pilot shows a significant rise in cheating with capability (o3 > o4-mini, Holm-robust) and an AS-vs-GP advantage that is directional but below the resolution the 103-task census affords; it is a bounded-null risk that no amount of compute fixes, because the pre-registered MID is at the substrate's detectability floor. Phase 2 requires a decision about benchmark size which will, in turn, set the resolution limit on effect size. Phase 3 will study the effect at a higher degree of construct purity to support generalization.

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Project references:

Spec-variant text: [src/pants_on_fire_eval/task.py](#).

Pre-registration: [docs/preregistration.md](#).

Statistical methodology: [docs/statistical-methodology.md](#).

Reproduction and substrate notes: [docs/step1-pilot.md](#).